Course Path

STAT 109 — Summer Session 1, 2026 (online, async)

This page shows the recommended path through the course during the five-week summer session from June 1 to July 2, 2026. You may work at your own pace within each module window, but the modules should be done in the order shown. Check Canvas to turn in work, take quizzes, see grades, and join discussions.

Each project module follows the same pattern: **Read 0** (project preview) → read → update your reference sheets → low-stakes quizzes (as many times as you want) → lab (Wednesday) → synthesis quiz (Thursday) → project milestone (Friday) → project (Sunday). Module 5 is shorter and ends on Thursday, July 2.

Reference sheets (methods map, R reference) grow a little in every module 1–5. All readings also live on the lectures index if you want the full list.

Outcomes below are tagged **essential** (needed for the module project or core module goals) or **optional** (helpful enrichment or stretch).

The **Supplemental (optional)** column links extra **textbook chapters** (Vu & Harrington, free online) and **videos/tutorials** that match our Colab + `{tidyverse}` workflow — see Resources. None of these are required for the module project unless a row says otherwise.

At a glance

Module	Dates	Project due
0 — Welcome	Open before Jun 1	—
1 — Testing a Claim	Jun 1–7	Sun Jun 7
2 — Estimating the Truth	Jun 8–14	Sun Jun 14
3 — Discovering Patterns	Jun 15–21	Sun Jun 21
4 — Designing Fair Comparisons	Jun 22–28	Sun Jun 28
5 — Looking Ahead	Jun 29–Jul 2	Thu Jul 2

Module 0 — Start here

Complete this before Module 1

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 0 Read 0	—	**Essential:** Welcome — five-step statistical process, three worlds, five portfolio projects, Colab/R notebook structure, and Module 0 checklist (labs, quizzes, research questions). ~15 min before the syllabus and labs.
Read	Syllabus, course path, resources	—	**Essential:** Know where readings, labs, and projects live vs what happens in Canvas. **Essential:** Understand grading, due-date pattern, and academic honesty / AI expectations (syllabus). **Optional:** Full policy details you will not need until later in the term.

Step	Materials	Supplemental (optional)	Learning outcomes
Quiz 0 (Start Here)	Canvas Module 0 — Quiz 0	—	**Essential:** Confirm Read 0 concepts: statistics definition, five-step process (ask → plan → collect → analyze → conclude), three worlds, Colab text/code/output, and basic R from the Module 0 lab. Open notes; unlimited attempts.
Quiz 0B (Question Ideas)	Canvas Module 0 — Quiz 0B	—	**Essential:** Submit 3 research questions for Step 1 practice; the third should compare two groups (Project 1 preview).
Reference sheets	Methods map, R reference	—	**Essential:** Set up two places to create your two references over the next 5 weeks. You will update these each module and can use them on assignments. **Optional:** Skim the methods map layout and R reference before Module 1 labs.
Lab (Wed)	Demo lab, Exercise lab	—	**Essential:** Run R in Google Colab (save a copy, run cells). **Essential:** Work with vectors, logical TRUE/FALSE, and simple simulation. **Optional:** Estimating probabilities by simulation (helps later probability intuition, not required for Project 1).
Canvas	Module 0 discussion	—	**Essential:** Introduce yourself and confirm you can reach assignments and discussions.
Supplemental pack	—	Colab tutorial · Posit: Getting Started with R · Textbook (free PDF) · Posit cheatsheets	**Optional:** Colab orientation and R basics before Module 1; textbook preface for context.

Module 1 — Testing a Claim (Jun 1–7)

Is this observation surprising?

Lab Wed Jun 3 · Synthesis Thu Jun 4 · Milestone Fri Jun 5 · Project 1 — Testing a Claim Sun Jun 7 · Discussion: something surprising about Lumberjacks (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 1 Read 0	—	**Essential:** Project preview — big question, sample ESP report (PDF), what to notice in each section, link to Project 1. ~5 min before Read 1.

Step	Materials	Supplemental (optional)	Learning outcomes
Read	Module 1 Read 1	—	**Essential:** Probability language through binomial pmf — sample spaces, rules, counting, and the binomial process.
Quizzes (Def & Notation)	Canvas Module 1 — Quiz A	—	**Essential:** Notation, sample space, addition rule, conditional probability, independence. Open notes; unlimited attempts.
Read more	Module 1 Read 2	—	**Essential:** Binomial test workflow and R — hypotheses, null plots, p-values.
Quizzes (Concept Check)	Canvas Module 1 — Quiz B	—	**Essential:** Counting, conditional applications, combinations, BRP identification. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	—	**Essential:** Probability vocabulary (sample space through independence) on statistics sheet. **Essential:** One-proportion / binomial test row on methods map. **Essential:** R patterns for `dbinom`, `pbinom`, `geom_col`, `geom_vline`.
Lab (Wed)	Demo lab, Conditional demo, Exercise lab, Project 1 milestone	—	**Essential (demos):** simulation review, data structures, conditional simulation, binomial R tools, null plot. **Essential (exercise):** combined practice notebook (Parts I–III). **Essential (milestone):** Project 1 precheck. Update R reference.
Synthesis quiz (Thu)	Canvas Module 1 — Synthesis	—	**Essential:** Full binomial test workflow, Module 1 R practice, interpretation items. After labs. Two attempts; open notes.
Milestone (Fri)	Project 1 milestone	—	**Essential:** Phase A — question, p, H_0/H_a, null graph **before** data collection.
Project (Sun)	Project 1 — Testing a Claim	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 2 — Probability · Ch 3.2 — Binomial · Khan: binomial distribution · Khan: visualizing binomial	**Optional:** Alternate probability explanations; course labs use `{ggplot2}` for null plots, not the textbook's base-R labs.

Project 1 — Testing a Claim

Outcome	Tag
Choose a real binomial process ($n = 25$) that is not a class example; defend all four BRP conditions in your own context	**Essential**
State H_0 and H_a for population proportion p with a numeric null value	**Essential**

Outcome	Tag
Report x, n, $\hat{p}$; descriptive bar/pie chart in R	**Essential**
Plot null binomial distribution with `dbinom` + `geom_col`; mark observed x	**Essential**
Compute p-value with `pbinom` matching alternative direction; interpret in one sentence	**Essential**
One-page report with Introduction / Data Collection / Data Analysis / Conclusion	**Essential**
Use the Read 2 ESP worksheet as a model without copying ESP as your study	**Essential**

Module 2 — Estimating the Truth (Jun 8–14)

Building high-quality estimates from samples.

Lab Wed Jun 10 · Synthesis Thu Jun 11 · Milestone Fri Jun 12 · Project 2 — Estimating the Truth Sun Jun 14 · Discussion: a plant you appreciate (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 2 Read 0	—	**Essential:** Project preview — finite population, SRS, 95% CI focus; tour sample report (PDF). ~5 min before Read 1.
Read	Module 2 Read 1	—	**Essential:** Binomial $\mu = np$, $\sigma = \sqrt{np(1-p)}$; normal approximation & Empirical Rule; sampling distribution & 95% CI (proportion and mean); sampling frame, bias, precision, SRS and design types.
Quizzes (Def & Notation)	Canvas Module 2 — Quiz A	—	**Essential:** Vocabulary for estimation and sampling (frame, SRS, bias, CI language). Open notes; unlimited attempts.
Read more	Module 2 Read 2	Textbook Ch 4.2 — Confidence intervals	**Essential:** z-scores & standard normal; methods-map habit; mean/median, histogram, dot plot in tidyverse. **Optional:** `prop.test()` / one-proportion z-test; one-sample `t.test()` (Project 2 uses CI only).
Quizzes (Concept Check)	Canvas Module 2 — Quiz B	—	**Essential:** z-scores, toolbox organization, one-variable summaries. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	—	**Essential:** One-mean and one-proportion CI rows; FPC note for finite populations. **Essential:** z-score row before toolbox section.

Step	Materials	Supplemental (optional)	Learning outcomes
Lab (Wed)	Module 2 demo, Exercise lab	—	**Essential:** Choose among CI situations (mean vs proportion; finite N; FPC). **Essential:** Check conditions and compute 95% CI in R.
Synthesis quiz (Thu)	Canvas Module 2 — Synthesis	—	**Essential:** CI computation and interpretation; SRS vs convenience; FPC decision; exercise-lab R patterns; tidyverse summaries. After labs.
Milestone (Fri)	Canvas Module 2	—	**Essential:** Named finite population (N), variable type, planned SRS method, and which CI type you will use.
Project (Sun)	Project 2 — Estimating the Truth	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 3.3 — Normal · Ch 4.1–4.2 — Variability & CIs · Khan: confidence intervals · Khan: CI simulation	**Optional:** CI intuition and normal-model background; Project 2 rubric is CI-focused.

Project 2 — Estimating the Truth

Outcome	Tag
Define a listable finite population (N) and one variable (binary $\rightarrow$ proportion, or numerical $\rightarrow$ mean)	**Essential**
Draw an SRS of size n from $\{1, \ldots, N\}$ without replacement; collect data	**Essential**
Compute point estimate ($\hat{p}$ or $\bar{x}$)	**Essential**
Choose correct 95% CI (mean t-interval or proportion interval; apply FPC when N is finite)	**Essential**
State and check interval conditions in context	**Essential**
Interpret CI with the "I am 95% confident that…" sentence	**Essential**
Submit one-page summary + runnable R notebook (mean or proportion template)	**Essential**
Sample size calculation / professional power analysis	**Optional**
Hypothesis test about the parameter (not required by rubric)	**Optional**

Module 3 — Discovering Patterns (Jun 15–21)

Describing a dataset with graphs and summaries.

Lab Wed Jun 17 · Synthesis Thu Jun 18 · Milestone Fri Jun 19 · Project 3 — Discovering Patterns Sun Jun 21 · Discussion: a place from home (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 3 Read 0	—	**Essential:** Project preview — EDA goals, variable-type table, sample Project 3 tour. ~5 min before Read 1.
Read	Module 3 Read 1	—	**Essential:** Compare numerical outcome across two groups with spread (range, IQR, SD), boxplots, and `group_by()` + `summarize()`. **Optional:** Welch two-sample t-test (Project 3 is descriptive only).
Quizzes (Def & Notation)	Canvas Module 3 — Quiz A	—	**Essential:** Tidyverse pipeline verbs (`filter`, `select`, `mutate`, `count`, `group_by`, `summarize`). **Essential:** Variable types (numerical vs categorical). **Essential:** Spread vocabulary (range, quartiles, IQR, SD, five-number summary). Open notes; unlimited attempts.
Read more	Module 3 Read 2	—	**Essential:** Scatterplots for two numerical variables; form, direction, strength. **Essential:** Two-way tables (joint/row/column proportions) and stacked bar charts.
Quizzes (Concept Check)	Canvas Module 3 — Quiz B	—	**Essential:** Choose the right graph/summary by variable type. **Essential:** Association description. **Essential:** Two-way table reading. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	—	**Essential:** Describe rows for one categorical, one numerical, two categorical, two numerical, and numerical-by-group. **Essential:** `ggplot` patterns for histogram, boxplot, bar, scatter, stacked bars. **Optional:** Inferential rows for two-group tests (not required for Project 3).
Lab (Wed)	Module 3 demo, Exercise lab	—	**Essential:** Five EDA situations: univariate numerical/categorical; boxplots by group; scatterplot; two-way table + stacked bars. **Essential:** `group_by` + `summarize`; `count` + `pivot_wider`.
Synthesis quiz (Thu)	Canvas Module 3 — Synthesis	—	**Essential:** Five-number summary; compare groups from boxplots; scatterplot description; two-way proportions; ggplot2 calls. After labs. Two attempts; open notes.
Milestone (Fri)	Canvas Module 3	—	**Essential:** Chosen dataset link, four variables with types, and plan for two figures + four numeric facts.

Step	Materials	Supplemental (optional)	Learning outcomes
Project (Sun)	Project 3 — Discovering Patterns	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 1 — Intro to data (§1.4–1.7) · R4DS: Data visualization · Posit: ggplot2 video	**Optional:** EDA and `{ggplot2}` depth; use course Colab labs for `{tidyverse}` practice (not the textbook's base-R lab sets).

Project 3 — Discovering Patterns

Outcome	Tag
Choose a dataset (TidyTuesday or other) with 4 variables and 10 rows; document source	**Essential**
Classify four variables as numerical or categorical	**Essential**
Univariate summaries and plots for each of the four variables	**Essential**
Bivariate exploration for variable pairs (tables/plots appropriate to types)	**Essential**
One-page PDF: context, two best graphics, 4 labeled numeric summaries, four findings	**Essential**
Notebook runs top-to-bottom with all code used (4 univariate + 6 bivariate plots in notebook; only 2 plots on one-pager)	**Essential**
Part A short answers in Canvas	**Essential**
Inferential tests or population generalization beyond the dataset	**Optional** (explicitly out of scope — descriptive statistics only)
Your own research question beyond exploratory findings	**Optional**

Module 4 — Designing Fair Comparisons (Jun 22–28)

Comparing two groups responsibly.

Lab Wed Jun 24 · Synthesis Thu Jun 25 · Milestone Fri Jun 26 · Project 4 — Designing Fair Comparisons Sun Jun 28 · Discussion: a place in Humboldt (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 4 Read 0	—	**Essential:** Project 4 preview; what makes a two-group comparison fair; vocabulary overview.
Read 1	Module 4 Read 1	—	**Essential:** EV vs RV; observational studies vs experiments; confounding; DAG with one confounder; "by chance" vs causal language. **Essential:** RCT definition; what random assignment buys. **Optional:** Adjustment sets / backdoor criterion beyond one confounder.

Step	Materials	Supplemental (optional)	Learning outcomes
Reference sheets	Methods, R reference	—	**Essential:** Two-group inferential rows (Welch t, two-proportion / χ^2 as appropriate). **Essential:** Study-design vocabulary. **Essential:** R pipeline: `group_by`, plots by group, `t.test` / `prop.test` / `chisq.test`.
Quiz A (Def & Notation)	Canvas Module 4 — Quiz A	—	**Essential:** RCT vs observational; confounder; DAG; causal vs association language.
Read 2	Module 4 Read 2	—	**Essential:** Pick two-group test from methods table; Welch t-test; two-proportion z-test; conditions; connect random assignment to reduced confounding.
Quiz B (Concept Check)	Canvas Module 4 — Quiz B	—	**Essential:** Design classification; methods-map matching; read two-group R output.
Demo lab	Module 4 demo lab	—	**Essential:** `set.seed()`; numerical outcome → Welch t-test; binary outcome → two-proportion z-test; validate under null; RCT vs observational confounding demo.
Exercise lab (submit)	Module 4 exercise lab	—	**Essential:** Four problems — same structure as demo. Connect simulation plan to Project 4 design.
Synthesis quiz	Canvas Module 4 — Synthesis	—	**Essential:** Design + methods table integration; interpret two-group CIs; causal vs association language; simulation R workflow; error types in context.
Milestone	Canvas Module 4	—	**Essential:** Assigned research question; draft answers to Part 1 design questions; plan for simulated data structure.
Project (Sun)	Project 4 — Designing Fair Comparisons	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 5.3 — Two-sample tests · Ch 8.1–8.2 — Proportions · R4DS: Data transformation	**Optional:** Two-group inference concepts; course labs use `group_by()`, `{ggplot2}`, and Colab.

Project 4 — Designing Fair Comparisons

Outcome	Tag
Answer ten study-design questions (EV/RV, operational definitions, hypotheses, RCT vs observational, confounder + DAG, test name from methods table)	**Essential**
Simulate reproducible dataset (`set.seed`) with binary EV, response, and confounder when needed	**Essential**

Outcome	Tag
Descriptive summaries and plots by group (numerical → boxplot/summaries; binary → stacked bars / proportions)	**Essential**
Run appropriate two-group test / CI; report p-value and interval; discuss "by chance" explanation	**Essential**
One-page summary written for the classmate who asked the question	**Essential**
Runnable notebook with simulation + analysis sections	**Essential**
Real-world data collection for the assigned question	**Optional** (project uses simulation)
Adjust for confounding in the inferential model	**Optional** (confounder included in design/DAG and plots; test does not remove confounding)

Module 5 — Looking Ahead (Jun 29–Jul 2)

Using data to make predictions.

This module is four days. Lab Wed Jul 1 · Project 5 — Looking Ahead, synthesis quiz, milestone, and portfolio package due Thu Jul 2 · Discussion: predicting a future observation (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 5 Read 0	—	**Essential:** Project preview — nesting dolls or stones; four-day schedule; what to expect in Read 1–2 and the lab. ~5 min before Read 1.
Read	Module 5 Read 1	Regression line applet	**Essential:** Scatterplot description (form, direction, strength); Pearson r; fit `lm(y ~ x)`; interpret slope and intercept; point prediction; residuals; MSE and R^2 on test set.
Quizzes (Def & Notation)	Canvas Module 5 — Quiz A	—	**Essential:** Scatterplot vocabulary, regression notation, slope/intercept interpretation, residual definition, R^2. Open notes; unlimited attempts.
Read more	Module 5 Read 2	Penn State: prediction intervals	**Essential:** Residual normality condition; 95% CI for slope with `confint()`; CI for mean response vs prediction interval with `predict(..., interval=)`; why PI is wider. **Optional:** Textbook Ch 6.5 on interval estimates with regression.
Quizzes (Concept Check)	Canvas Module 5 — Quiz B	—	**Essential:** Residual conditions, CI for slope, mean-response CI vs prediction interval. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	—	**Essential:** Two-numerical describe + regression / prediction rows. **Essential:** `lm`, `predict`, scatterplot with fitted line on R sheet.

Step	Materials	Supplemental (optional)	Learning outcomes
Lab (Wed)	Demo lab, Exercise lab	—	**Essential (demo):** Scatterplot + `cor`; fit `lm`; point prediction; `confint` for slope; `predict` with CI and PI. **Essential (exercise):** Apply same five-step workflow to your team's Project 5 data.
Synthesis quiz (Thu)	Canvas Module 5 — Synthesis	—	**Essential:** Read scatterplots; interpret r, slope, intercept; prediction vs confidence interval; read `predict` output; portfolio method review. After labs. Two attempts; open notes.
Milestone (Thu)	Canvas Module 5	—	**Essential:** Team, option (nesting dolls or stones), variables defined, data collection underway.
Project (Thu)	Project 5 — Looking Ahead	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 6 — Simple linear regression · Khan: regression line example · Penn State: prediction intervals	**Optional:** Regression concepts and PI vs CI; Project 5 requires SLR only.

Project 5 — Looking Ahead

Outcome	Tag
Form a team (1–4); choose Option A (nesting dolls) or B (stone container)	**Essential**
Define measurement rules and collect team dataset (CSV / shared sheet)	**Essential**
Fit at least one simple linear model `lm(response ~ predictor)`	**Essential**
Report prediction equation, plugged prediction for target case, 95% prediction interval	**Essential**
One-page PDF: intro, methods, results, conclusion, team contributions	**Essential**
Runnable team `.ipynb` from data load through model and prediction	**Essential**
Compare multiple models / predictors and choose best	**Optional** (encouraged but one SLR minimum)
Multiple predictors / logistic regression for binary outcomes	**Optional** (beyond Project 5 scope)
Portfolio package: both reference sheets + all five projects polished for submission	**Essential**

Notes

- **Supplemental (optional)** links point to the Vu & Harrington textbook and Colab/`{tidyverse}`-friendly videos — not required for the module project. The textbook's companion R labs use base R; prefer this course's Colab notebooks and Resources.
- Modules unlock in order in Canvas (0 → 1 → … → 5).
- Quizzes are open notes: use your reference sheets and the methods map.
- Definitions & Notation and Concept Check: unlimited attempts. Synthesis: two attempts (highest score counts).

- The milestone is an ungraded checkpoint — it helps you and me catch problems before the project is due.
- Modules 1–4: lab Wednesday, synthesis Thursday, milestone Friday, project Sunday.
- Module 5: lab Wednesday July 1; synthesis, milestone, project, and portfolio package due Thursday July 2.
- Quiz source files (`quizzes/quiz-*.qmd`) are practice banks; Canvas quizzes may use a subset. Items marked **optional** above may still appear — they are not required for the module project.
- If dates shift when the term opens, Canvas is the official calendar; this page will be updated to match.